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Unit 3: Computer Networks

An Interactive Learning Journey for 10th Class Computer Science

A Note Before We Begin

You are about to explore one of the most powerful inventions in human history — the computer network. Every time you send a message, watch a video, or search for something online, a network is working silently behind the scenes. By the end of this chapter, you will understand exactly *how* it all works. Some ideas might feel confusing at first. That is perfectly normal. Even professional engineers took time to understand these things. Take a breath, read slowly, and enjoy the journey.

Student Learning Outcomes

By the end of this chapter, you will be able to:

- Explain what a computer network is and why we use it.
 - Describe the key parts of data communication.
 - Identify networking devices like switches, routers, and access points.
 - Compare different network topologies (arrangements of devices).
 - Explain the three modes of data transmission.
 - Describe the 7-layer OSI model and what each layer does.
 - Understand the difference between IPv4 and IPv6 addresses.
 - Explain what protocols are and how DNS, DHCP, and TCP/IP work.
 - Describe the importance of network security and common security methods.
 - Identify types of networks: PAN, LAN, MAN, WAN, and CAN.
 - Give real-world examples of networks in business, education, and healthcare.
-

3.1 Network as a System

The Hook: Two Letters That Changed the World

It was October 29, 1969. A student at UCLA named Charley Kline sat down at a computer terminal. He was going to send the very first message over a new network called **ARPANET** — the great-great-grandfather of today's Internet.

The message was simple: the word "**LOGIN**".

He typed the letter "**L**". He called Stanford University on the phone to check if they received it. They had.

He typed the letter "**O**". They received it.

He typed the letter "**G**" — and the entire system **crashed**.

The very first message ever sent over a computer network was just two letters: "**LO**".

Today, billions of messages travel across networks every second, without crashing. But it all started with those two fragile letters. This is the world of computer networks — a world built piece by piece, error by error, into something extraordinary.

What is a Computer Network?

A **computer network** is a system of two or more devices that are connected together so they can **share information and resources**.

Think of it like a road system. Roads connect cities. A computer network connects devices — computers, phones, printers, and more.

Networks can be small or large:

- A **small network** might connect just two computers in a house.
- A **large network** is the **Internet** — connecting billions of devices all over the world.

The Internet is so large that it is often called the "**network of networks**" — because it connects thousands of smaller networks together.

Primary Components of a Network

Every network is made of a few key building blocks. Here they are:

Component	What it is	Simple Example
Node	Any device connected to the network	Your laptop, phone, or printer
Link	The connection between nodes	An Ethernet cable or a Wi-Fi signal
Switch	A device that connects multiple nodes inside one network	The device in your school lab connecting all computers
Router	A device that connects different networks together	The box at home that gives you the Internet

Objectives of Computer Networks

Why do we build networks in the first place? There are three main goals:

1. Resource Sharing

Imagine a school with 30 computers but only 2 printers. Without a network, each computer would need its own printer. That would be very expensive! With a network, all 30 computers can **share** those 2 printers.

Resource sharing means connecting devices so they can use each other's hardware, files, and software — saving time and money.

Real-Life Example: In an office, all employees share one large printer through the network instead of having a separate printer on every desk.

2. Data Communication

Networks allow people to **send and receive information** with each other, no matter where they are.

This includes:

- Sending **emails**
- Making **video calls** (like Zoom or Google Meet)
- Chatting via **instant messaging** (like WhatsApp)

Real-Life Example: A teacher in Lahore can video call a student in Karachi using an internet connection. The network carries their voices and faces across hundreds of kilometres in real time.

3. Connectivity and Collaboration

Networks connect people so they can **work together** even when they are far apart.

Real-Life Example: A team of students can all edit the same Google Doc at the same time, from different homes, using the internet.

Pause & Think — 3.1

Scenario: Your school has 20 computers in the computer lab. The school wants to save money. Currently, each computer has its own printer.

Question: If the school connects all 20 computers into one network, how many printers might it need instead of 20? What is the benefit? What could go wrong if the network is slow?

Discuss your answer with a partner before reading on.

3.2 Fundamental Concepts in Data Communication

The Hook: A Letter in Pieces

Imagine you want to send a very long letter to your friend who lives far away. But the postal office has a rule: every envelope can only hold **one page**. So you tear your letter into pages, put each page in a different envelope, and send them all.

Your friend receives all the envelopes — maybe in a different order — and puts the pages back together to read your message.

This is almost exactly how **data communication** works on a computer network. Your message is broken into small pieces, sent separately, and reassembled at the other end. The system that governs all of this is called **data communication**.

What is Data Communication?

Data communication is the process of sending data (information) from one device to another through a connection.

Every act of data communication has **five essential components**. Think of them like the five things needed to send a letter:

The Five Components of Data Communication

1. Sender

The **sender** is the device or person that *sends* the data.

 **Example:** Your smartphone when you send a WhatsApp message.

2. Receiver

The **receiver** is the device or person that *receives* the data.

Example: Your friend's phone that receives your WhatsApp message.

3. Message

The **message** is the actual data being sent. It can be text, a photo, a video, a voice note — anything.

Example: The text "Hello! Are you free?" in your WhatsApp message.

4. Protocol

A **protocol** is a **set of rules** that both the sender and receiver agree to follow. Without shared rules, the receiver would not understand the message.

Think of a protocol like the rules of a language. If you write a letter in Urdu, the receiver must also know Urdu to read it. Both sides must follow the same rules.

Example: When you visit a website, your browser and the website follow a protocol called **HTTP** (HyperText Transfer Protocol) to communicate correctly.

5. Medium

The **medium** is the *path* through which data travels from sender to receiver.

There are two types of medium:

- **Wired medium:** Data travels through physical cables (like Ethernet cables).
- **Wireless medium:** Data travels through radio waves or light (like Wi-Fi or Bluetooth).

Example: When you connect a computer to the internet using a cable, the cable is the medium. When you connect using Wi-Fi, the radio waves are the medium.

Quick Visual Summary

```
[ SENDER ] ---[ MESSAGE ]--->---[ MEDIUM ]--->---[ RECEIVER ]
                                |
                                [ PROTOCOL ]
                        (rules both sides must follow)
```

Pause & Think — 3.2

Scenario: You are sending a photo from your phone to your friend's phone using WhatsApp over Wi-Fi.

Challenge: Can you identify all five components of data communication in this scenario?

- What is the sender?
- What is the receiver?
- What is the message?
- What is the medium?
- What might the protocol be?

Write your answers before checking the section above.

3.3 Networking Devices

The Hook: The Post Office Problem

Imagine a city where every person can only deliver letters directly to another person. With 10 people, that works fine. But with 10,000 people, it becomes chaos — everyone running in every direction.

So we built **post offices**: central places where letters are sorted and sent to the right destination. Computer networks have their own version of post offices — they are called **networking devices**: switches, routers, and access points.

□ Device 1: The Switch

What is a Switch?

A **switch** is a networking device that connects multiple devices *within the same network* (for example, all the computers in one school lab) and makes sure data reaches the correct device.

A switch is like a **smart traffic conductor** inside a building. It reads the destination on each data packet and sends it to *exactly* the right device — not to everyone.

How Does a Switch Work?

Every device on a network has a unique ID called a **MAC Address** (Media Access Control address). Think of a MAC address like a permanent fingerprint for your device. It never changes.

When data arrives at the switch, the switch reads the MAC address (the destination fingerprint) and sends the data *only* to the correct device.

Step-by-Step: How a Switch Sends a File

Let's say you want to send a file from your computer (Computer A) to your classmate's computer (Computer B) in the same lab.

1. **You click "Send."** Your computer breaks the file into small pieces called **packets**.
2. **Each packet is labelled.** Each packet contains the destination MAC address (Computer B's fingerprint).
3. **Packets travel to the switch.** All packets arrive at the network switch.
4. **The switch reads the MAC address.** It looks at the destination fingerprint on each packet.

5. **The switch sends packets to Computer B only.** No other computer receives these packets.
6. **Computer B receives all packets and reassembles them** into the original file.

Important Note: The very first time a switch sees a device, it does not yet know its MAC address, so it sends the data to *all* devices (this is called **broadcasting**). After it learns the addresses, it becomes efficient and sends data only to the correct device.

Key Facts About Switches

- Switches work at **Layer 2** (the Data Link Layer) of the OSI model (we will cover this in Section 3.6).
- Switches use **MAC addresses** to deliver data.
- Switches connect devices **within** a single network.
- A switch makes the network **faster** because it doesn't send data to everyone — only to the right device.

Device 2: The Router

What is a Router?

A **router** is a networking device that connects *different networks* together and directs data between them.

If a switch is like a traffic conductor *inside* a building, a **router** is like the traffic officer at the city intersection — deciding which *road* to use to get your data to a *completely different city* (a different network).

Real-Life Analogy: Think of air travel. If you want to fly from Lahore to London, you might stop at an airport in Dubai. The airports are like routers — they direct you along the best path to your final destination.

How Does a Router Work?

When data is sent across the internet, it is broken into **packets**. Each packet contains:

1. The actual data (a piece of your message or file)
2. The **IP address** of the destination (a unique address identifying *which network* and *which device* should receive it — more on this in Section 3.7)

The router looks at the IP address on each packet and decides the **best path** to send it. Routers use a special table called a **routing table** — a list of known paths and networks — to make this decision.

Step-by-Step: How a Router Sends Data to the Internet

1. **You type a website address** in your browser (e.g., www.google.com).
 2. **Your device sends a request.** The request is broken into packets, each labelled with Google's IP address.
 3. **Packets arrive at your home router.** Your router reads the destination IP address.
 4. **Your router forwards packets** toward the internet, passing them through other routers.
 5. **Each router along the path** reads the destination IP and forwards packets closer to the destination.
 6. **Google's server receives the packets**, reassembles them, and sends back the webpage.
 7. **Your browser displays the webpage.**
-

Key Facts About Routers

- Routers work at **Layer 3** (the Network Layer) of the OSI model.
 - Routers use **IP addresses** to route data.
 - Routers connect **different networks** together (e.g., your home network to the internet).
 - Routers use a **routing table** to choose the best path for data.
-

Device 3: The Access Point

What is an Access Point?

An **Access Point (AP)** is a device that allows wireless devices (phones, laptops, tablets) to connect to a wired network.

Think of an access point like a **bridge**. On one side is a wired network (connected by cables). On the other side are your wireless devices. The access point allows wireless devices to cross the bridge and join the wired network.

Real-Life Example: The Wi-Fi router in your home acts as both a router *and* an access point. It connects to the internet via a cable, and then creates a wireless signal that your phone and laptop can connect to.

How Does an Access Point Work?

1. The access point receives data from the **wired network** through a cable.
2. It **converts** that data into **wireless radio signals**.
3. Your wireless devices (phone, laptop) **receive** those radio signals.
4. Your device **sends back** data as radio signals.
5. The access point **converts** those signals back into wired data and sends them along.

Tip: For the best Wi-Fi signal, place your access point in the **centre** of the area you want to cover. Walls and metal objects can weaken the signal.

Grab a Partner — 3.3

Role Play: One person plays a **switch**, the other plays a **router**.

- The switch person: Your job is to direct messages *within the classroom* (your local network). You only send messages to the specific student whose name is on the message.
- The router person: Your job is to decide which *door* (which path out of the classroom) a message should use if it needs to go to another classroom (a different network).

Now act it out: One student writes "Send this to Ahmed in Room 5" on a piece of paper. Decide together — does the switch or the router handle this? What if it said "Send this to Ahmed in another school building"?

3.4 Network Topologies

The Hook: Classroom Desk Arrangements

Have you ever noticed how your teacher arranges desks differently for different activities? Sometimes all desks face the front (like a lecture). Sometimes desks are arranged in a circle (for discussion). Sometimes desks are arranged in small clusters (for group work).

Each arrangement has advantages and disadvantages. A circle is great for discussion but bad if the teacher needs to write on the board. Small clusters are great for teamwork but hard to monitor.

Network topology is exactly the same idea — it is the *arrangement* of devices in a network. The way you arrange devices affects speed, reliability, and cost.

What is Network Topology?

Network topology refers to the *physical or logical arrangement* of devices (nodes) and connections (links) in a computer network.

There are four main topologies you need to know:

Topology 1: Bus Topology

What is Bus Topology?

In a **bus topology**, all devices are connected to **one single central cable**, called the **bus** or **backbone**. Data travels along this cable in both directions.

```
[Computer 1]---[Computer 2]---[Computer 3]---[Computer 4]
|-----|
BUS CABLE
```

Analogy: Imagine a single road (the bus cable) passing through a village. Every house (computer) has a driveway connecting to this one road. If the road is blocked anywhere, no one can travel.

Advantages of Bus Topology

- ✗ • **Simple to set up.** You just connect devices to one cable.
- **Low cost.** You need less cable than other topologies.
- **Good for small networks** with few devices.

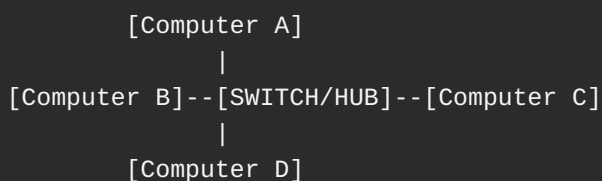
Disadvantages of Bus Topology

- **If the main cable fails, the entire network goes down.** Every device loses connection.
 - **Traffic jams.** If many devices send data at the same time, they can interfere with each other.
 - **Hard to identify problems.** When something goes wrong, it is difficult to find where.
-

Topology 2: Star Topology

What is Star Topology?

In a **star topology**, every device is connected to **one central device** — usually a switch or a hub. All data passes through this central device.



Analogy: Think of a school principal's office connected to every classroom by a separate intercom line. All communication goes through the principal's office. If the office is closed, no one can communicate.

Advantages of Star Topology

- ✗ • **Easy to identify problems.** If one connection fails, only that one device is affected.
- **Easy to add new devices.** Just connect them to the central switch.
- **Better performance.** Each device has its own connection to the switch, reducing traffic jams.

Disadvantages of Star Topology

- **If the central switch fails, the entire network goes down.** This is the single weak point.
- **More cable is needed.** Each device needs its own cable to the centre.
- **More expensive** than bus topology because of the central switch and extra cables.

Most modern school labs and offices use **star topology** because it is reliable and easy to manage.

Topology 3: Ring Topology

What is Ring Topology?

In a **ring topology**, every device is connected to exactly **two other devices**, forming a **closed circle**. Data travels in one direction around the ring, passing through each device until it reaches its destination.

```
[Computer A]
  |         \
[Computer D] [Computer B]
  |         /
[Computer C]
```

Analogy: Imagine a relay race. Each runner passes the baton to the next runner in a circle. The baton keeps going around the ring until it reaches the right person.

Advantages of Ring Topology

- **Equal access for all devices.** No one device hogs the network.
- **Can handle heavy traffic.** Works well when many devices are sending data.
- **Easy to identify which device is causing a problem.**

Disadvantages of Ring Topology

- **If one connection breaks, the whole network is affected.** The ring is broken.
- **Adding a new device is disruptive.** You must interrupt the network to insert a new device into the ring.
- **Data moves slower** because it must pass through every device on the way.

Note: A **two-way ring** (where data can travel in both directions) solves many of these problems. If one path breaks, data uses the other direction.

Topology 4: Mesh Topology

What is Mesh Topology?

In a **mesh topology**, every device is connected directly to every other device. This creates many different paths for data to travel.

```
[A] --- [B]
 | \ / |
 |  X  |
 | / \ |
 [C] --- [D]
```

Analogy: Imagine a city where every house has a direct road to every other house. If one road is closed, you can always take a different road to reach your destination.

Advantages of Mesh Topology

- **Very reliable.** If one connection fails, data can take a different path.
- **High performance.** Data can take the shortest, fastest path.
- **No single point of failure.** The network keeps working even when parts break.

Disadvantages of Mesh Topology

- **Very expensive.** You need many cables and connections.
- **Difficult to set up and manage.** With many connections, it is complex.
- **Not practical for large networks** — the number of cables grows enormously as you add more devices.

Where is mesh used? The Internet itself uses a form of mesh topology, which is why it keeps working even when many cables and routers fail.

Quick Comparison Table

Topology	Central Device?	If one link fails	Cost	Best For
Bus	No	Whole network fails	Low	Tiny networks
Star	Yes (switch/hub)	Only one device affected	Medium	Schools, offices
Ring	No	Whole network affected	Medium	Specific industrial uses
Mesh	No	Network still works	High	Critical systems, internet backbone

Pause & Think — 3.4

Scenario: You are the IT manager of a new school. The school has 30 computers in one lab. The school's budget is limited, but reliability is important.

Challenge: Which topology would you choose — Bus, Star, Ring, or Mesh? Give **two reasons** for your choice and **one trade-off** (something you are giving up by choosing it).

There is no single right answer. Think it through!

3.5 Transmission Modes

The Hook: One-Way Streets and Highways

Imagine three types of roads:

- A **one-way street**: cars only go in one direction. No car comes back the same way.
- A **narrow lane**: cars can go in both directions, but only one at a time. One car must wait for the other to pass.
- A **two-lane highway**: cars travel in both directions at the same time, freely.

Data communication works the same way. The **transmission mode** describes how data flows between devices. There are three modes: **Simplex**, **Half-Duplex**, and **Full-Duplex**.

Mode 1: Simplex Communication

What is Simplex?

In **Simplex** communication, data flows in **only one direction**. The sender can only send, and the receiver can only receive. The roles never switch.

```
[ SENDER ] -----> [ RECEIVER ]
      (data flows one way only)
```

Real-Life Example: A keyboard sending keystrokes to a computer. The keyboard can only send. The computer receives. The keyboard never receives anything from the computer.

Another Example: A traditional television broadcast. The TV station sends signals to your TV. Your TV receives. Your TV cannot send anything back to the station.

✖ Advantages

- Simple and inexpensive.
- Useful when only one-way communication is needed.

Disadvantages

- No feedback is possible from receiver to sender.
- Not suitable when two-way communication is needed.

Mode 2: Half-Duplex Communication (one at a time)

What is Half-Duplex?

In **Half-Duplex** communication, data can flow in **both directions**, but **not at the same time**. One device sends while the other waits. Then they switch.

```
[ DEVICE A ] ———> [ DEVICE B ]  (A is sending, B is waiting)
[ DEVICE A ] <—— [ DEVICE B ]  (B is sending, A is waiting)
```

Real-Life Example: A walkie-talkie. When you press the button and talk, the other person must wait and listen. When you say "Over," you release the button and the other person can reply. Both cannot talk at the same time.

Advantages

- Both sides can communicate.
- Simpler than full-duplex because only one channel is needed.

Disadvantages

- Communication is slower because you must wait for your turn.
- Not efficient when both sides need to communicate simultaneously.

Mode 3: Full-Duplex Communication (simultaneously)

What is Full-Duplex?

In **Full-Duplex** communication, data flows in **both directions simultaneously**. Both devices can send and receive at the same time.

```
[ DEVICE A ] ———> [ DEVICE B ]
[ DEVICE A ] <—— [ DEVICE B ]
      (both happening at the same time)
```

Real-Life Example: A telephone call. Both people can talk and listen at the same time. You do not need to wait for the other person to stop speaking before you can speak.

Fun Fact: Early telephones were actually **half-duplex** — only one person could speak at a time! Modern digital phones are full-duplex.

Advantages

- Fast and efficient.
- Natural, like a face-to-face conversation.

Disadvantages

- More complex and expensive to implement.
- Requires more bandwidth (capacity for data).

Quick Comparison

Mode	Direction	Simultaneous?	Example
Simplex	One way only	N/A	Keyboard to computer
Half-Duplex	Both ways	No (take turns)	Walkie-talkie
Full-Duplex	Both ways	Yes	Phone call

Pause & Think — 3.5

Scenario: You are designing a communication system for a school's security guards. Guards need to contact each other and also receive messages from the main security office.

Challenge: Should you give them walkie-talkies (half-duplex) or smartphones (full-duplex)? Write down **two advantages** of your choice and **one situation** where the other option would actually be better.

3.6 The OSI Networking Model

The Hook: Sending a Letter the Long Way

When you send an international letter, many things happen:

1. You write the letter.
2. You put it in an envelope and write the address.
3. You put stamps on it.
4. A postal worker collects it.
5. It gets sorted at a local post office.
6. A truck takes it to the airport.
7. A plane carries it to another country.
8. It gets sorted again.
9. A postal worker delivers it.

Each of these steps is handled by a **different person or system** — and yet the letter arrives correctly. No one person does everything.

Computer networks work the same way. When you send data, it passes through **7 distinct layers**, each with a specific job. This system is called the **OSI Model**.

What is the OSI Model?

The **Open Systems Interconnection (OSI) Model** is a framework that describes how data travels from one device to another over a network. It was created so that different systems, from different manufacturers, could communicate with each other using the same set of rules.

The OSI model has **7 layers**. Each layer handles a specific part of the communication process.

Key Idea: When data is sent, it passes **down** through all 7 layers on the sender's side. When it is received, it passes **up** through all 7 layers on the receiver's side.

The 7 Layers of the OSI Model

Here is an easy way to remember the layers (from Layer 7 to Layer 1):

"All People Seem To Need Data Processing"

Application → Presentation → Session → Transport → Network → Data Link → Physical

Layer 7: Application Layer (The Closest to You)

What it does: This layer is where **you** interact with the network. It provides network services directly to the applications you use — email, web browsers, file transfer, etc.

Think of it as: A waiter in a restaurant. The waiter (Application Layer) takes your order (your request) and brings you what you asked for (the data).

Examples of protocols at this layer: HTTP (web browsing), SMTP (email sending), FTP (file transfer).

Layer 6: Presentation Layer (The Translator)

What it does: This layer **translates data** into a format that the application can understand. It also handles **encryption** (scrambling data so it is secure) and **compression** (making data smaller so it travels faster).

Think of it as: A translator converting a book from Arabic to English so more people can read it. The data is "translated" into a format the receiving application can understand.

Real-Life Connection: When you visit a website with "HTTPS," the Presentation Layer is handling the encryption that protects your data.

Layer 5: Session Layer (The Call Manager)

What it does: This layer **establishes, maintains, and terminates connections** (called sessions) between two devices. It makes sure a session does not drop accidentally.

Think of it as: The beginning, middle, and end of a phone call. Someone dials (session starts), you talk (session maintained), someone hangs up (session ends).

Example: When you log in to a website, a session is created. The session layer manages that connection for as long as you are logged in.

Layer 4: Transport Layer (The Delivery Service)

What it does: This layer ensures that data is delivered **completely and correctly** from source to destination. It breaks large data into smaller segments, numbers them, and reassembles them at the other end. It also handles **error checking**.

Think of it as: A courier delivery service. The service ensures your package arrives safely, completely, and in the right order.

Key Protocols:

- **TCP (Transmission Control Protocol):** Reliable delivery. Every packet is confirmed. Used for emails and web pages.
- **UDP (User Datagram Protocol):** Fast but no confirmation. Used for video calls and online gaming where speed matters more than perfection.

Layer 3: Network Layer (The GPS Navigator)

What it does: This layer is responsible for **routing** data between different networks. It determines the **best path** for data to travel from source to destination.

Think of it as: A GPS system. You enter your destination, and the GPS finds the best route across the city — through highways, turns, and shortcuts. The Network Layer does the same for data packets, routing them through the internet.

Key Protocol: IP (Internet Protocol) — this is where **IP addresses** are used to identify devices across different networks.

Layer 2: Data Link Layer (The Local Traffic Light)

What it does: This layer handles **node-to-node data transfer** within the same network. It ensures that data moves correctly from one device to the next on the same network, and detects and corrects errors from the Physical Layer.

Think of it as: Traffic lights at an intersection. Traffic lights manage the flow of cars (data) at local intersections (within one network) and prevent collisions.

Key Concept: The Data Link Layer uses **MAC addresses** to identify devices within a local network. This is where **switches** operate.

Layer 1: Physical Layer (The Actual Wires)

What it does: This layer is responsible for the actual **physical transmission** of raw data bits (0s and 1s) over a physical medium — cables, radio waves, fibre optic light, etc.

Think of it as: The actual road itself. Before any cars can drive, there must be a physical road. Before any data can travel, there must be a physical medium.

Examples: Ethernet cables, Wi-Fi radio signals, fibre optic cables, connectors, and hubs.

OSI Model Summary Table

Layer	Number	Name	Job	Example
Closest to user	7	Application	Provides network services to apps	HTTP, Email, FTP
	6	Presentation	Translates, encrypts, compresses data	HTTPS encryption
	5	Session	Manages connections (sessions)	Login sessions
	4	Transport	Reliable delivery, error checking	TCP, UDP
	3	Network	Routing between networks	IP addresses, Routers
	2	Data Link	Local data transfer, error detection	MAC addresses, Switches
Closest to hardware	1	Physical	Transmits raw bits	Cables, Wi-Fi signals

Pause & Think — 3.6

Scenario: You open your browser and type `www.school.edu`. The page loads in 2 seconds.

Challenge: Match each of the following events to an OSI layer:

1. The Wi-Fi signal carries your request wirelessly to the router.
2. The browser creates an HTTP request.
3. The router figures out which path to send your request.
4. The data is broken into numbered segments to ensure correct delivery.
5. Your login session to the school website is maintained as you browse.

Write the layer number (1–7) next to each event.

3.7 IPv4 and IPv6

The Hook: Running Out of Phone Numbers

In the early days of mobile phones, phone numbers were short — 5 or 6 digits. But as millions of people got phones, the world ran out of short numbers. Phone companies had to add more digits.

The same thing happened with internet addresses. When the internet was young, engineers created a system of addresses called **IPv4**. It seemed like enough — about 4 billion addresses. But by the 2000s, billions of devices were connecting to the internet — not just computers, but phones, TVs, refrigerators, cars. They ran out.

So engineers created a new system: **IPv6** — with addresses so numerous they could give a unique address to every atom on the surface of the Earth.

What is an IP Address?

An **IP address** (Internet Protocol address) is a **unique number** assigned to every device that connects to a network. It works like a home address — it tells the network exactly *where* to send data.

There are two versions of IP addresses: **IPv4** and **IPv6**.

IPv4: The Original System

IPv4 stands for **Internet Protocol version 4**. It is the fourth — and most widely used — version of the IP protocol.

Format of an IPv4 Address

An IPv4 address is **32 bits** long. It is written as **four groups of numbers**, each ranging from **0 to 255**, separated by dots.

```
Example: 192 . 168 . 1 . 1
          ↑     ↑     ↑     ↑
        Group1 Gr2  Gr3  Gr4
        (8 bits each = 32 bits total)
```

Think of it like: A home address with four parts — country, city, street, house number.

How Many IPv4 Addresses Are There?

The total number of possible IPv4 addresses is calculated as:

$2^{32} = 4,294,967,296$ — about **4.3 billion** addresses.

That sounds like a lot! But with over 8 billion people in the world, each with multiple devices, 4.3 billion was not enough.

IPv6: The New System

IPv6 stands for **Internet Protocol version 6**. It was created to solve the problem of running out of IPv4 addresses.

Format of an IPv6 Address

An IPv6 address is **128 bits** long. It is written as **eight groups of four hexadecimal digits** (numbers and letters), separated by colons.

```
Example: 2001 : 0000 : 130F : 0000 : 0000 : 0900 : 876A : 130B
```

This looks complicated! But don't worry — devices manage these addresses automatically. You don't need to memorise them.

How Many IPv6 Addresses Are There?

The total number of possible IPv6 addresses is:

$2^{128} = 340,282,366,920,938,463,463,374,607,431,768,211,456$

That is **340 undecillion** addresses — more addresses than there are grains of sand on every beach on Earth. We will not run out of IPv6 addresses for a very, very long time.

IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Length	32 bits	128 bits
Format	Four numbers (e.g., 192.168.1.1)	Eight groups with colons (e.g., 2001:0000:130F...)
Total addresses	~4.3 billion	~340 undecillion
Status	Still widely used	Gradually replacing IPv4
* Example	192.168.0.1	2001:0db8:85a3::8a2e:0370:7334

Practical Walkthrough: How to Find Your IPv4 Address on Windows

Here is a step-by-step guide to finding your own device's IP address. You can try this on any Windows computer in your school lab!

1. **Press the Windows key** on your keyboard. The Start menu opens.
2. **Type `cmd`** using your keyboard. You are searching for a program called "Command Prompt."
3. **Press Enter.** A black window opens. This is the Command Prompt.
4. **Type `ipconfig`** and press **Enter**.
- *What just happened?* The computer shows you all the network information for your device.
5. **Look for the line that says `IPv4 Address`.** Next to it, you will see your IP address — something like `192.168.1.5`.
6. **Look at each number.** Numbers range from 0 to 255. This is your device's unique address on the local network.

Note: The address you see is your **local network** address (within your school or home network). Your **public** internet IP address (the one websites see) may be different.

Pause & Think — 3.7

Scenario: Your school has 35 computers, 35 student phones, and 10 teacher devices — all connecting to the school Wi-Fi. The school IT manager says they are "moving from IPv4 to IPv6."

Question: Why might a school need IPv6? How many devices are we talking about? What would happen if two devices accidentally got the *same* IP address?

Discuss with a partner.

3.8 Network Protocols

The Hook: The UN Translator Problem

Imagine the United Nations General Assembly. Leaders from 193 countries arrive. Some speak Arabic, some speak French, some speak Mandarin, some speak Spanish. If everyone spoke at once in their own language, nothing would be understood.

So the UN uses **translators** and **rules**: only one person speaks at a time, you state your name before speaking, you use formal language. These rules allow people who speak different languages to communicate.

Network protocols are exactly these rules — they allow computers that may be made by different companies, running different software, in different countries, to communicate perfectly with each other.

What is a Protocol?

A **protocol** is a **set of agreed-upon rules** that govern how data is sent, received, and interpreted in a network.

Without protocols, computers could not understand each other. It would be like two people trying to have a conversation — one speaking Urdu and the other responding in Japanese.

Common protocols include:

- **HTTP / HTTPS** — for loading web pages
 - **FTP** — for transferring files
 - **SMTP** — for sending emails
 - **TCP/IP** — for all general internet communication
 - **DNS** — for translating website names to IP addresses
 - **DHCP** — for automatically assigning IP addresses to devices
-

DNS: The Internet's Phone Book

What is DNS?

DNS stands for **Domain Name System**.

When you type `www.google.com` in a browser, your computer needs to find the **IP address** of Google's server (because computers communicate using IP addresses, not names).

DNS is the system that **translates human-readable website names** (like `www.google.com`) **into IP addresses** (like `142.250.190.78`).

Analogy: DNS is like a phone book. You look up a person's name (the website name) and find their phone number (the IP address).

Step-by-Step: What Happens When You Type a Website Address?

1. **You type** `www.school.edu` in your browser and press Enter.
2. **Your device asks a DNS server:** "What is the IP address for `www.school.edu`?"
3. **The DNS server looks up the name** in its database.
4. **The DNS server replies** with the IP address, for example: `203.45.67.89`.
5. **Your browser uses the IP address** to contact the school's web server.
6. **The web server sends back the webpage.**
7. **Your browser displays the page.**

Key Point: This entire process happens in **milliseconds** — so fast that you never notice it.

DHCP: The Automatic Address Assigner

What is DHCP?

DHCP stands for **Dynamic Host Configuration Protocol**.

Every device on a network needs an IP address. But with many devices joining and leaving a network (like students' phones connecting to school Wi-Fi), assigning addresses *manually* would be impossible.

DHCP automatically assigns IP addresses to devices when they join the network — and takes the address back when they leave.

Analogy: Imagine a parking lot where an attendant assigns you a parking space when you arrive and frees it when you leave. DHCP is that attendant — automatically managing who gets which IP address at any given time.

Step-by-Step: How DHCP Works

1. **Your device connects to the Wi-Fi.**
 2. **Your device broadcasts a message:** "Hello! I just joined the network. Can someone give me an IP address?"
 3. **The DHCP server receives the message.** (The DHCP server is usually built into the router.)
 4. **The DHCP server assigns an available IP address** (e.g., `192.168.1.45`) to your device.
 5. **Your device uses this IP address** to communicate on the network.
 6. **When your device leaves**, the IP address is returned and can be given to another device.
-

TCP/IP: The Foundation of the Internet

What is TCP/IP?

TCP/IP stands for **Transmission Control Protocol / Internet Protocol**. It is the **fundamental suite (set) of protocols** that powers the entire internet.

It is actually made of two separate but related protocols:

TCP — Transmission Control Protocol

TCP ensures that data is delivered **reliably and completely**.

- It breaks data into **packets** (small pieces) and numbers them.
- It ensures every packet arrives at the destination.
- If a packet is lost, TCP asks for it to be sent again.
- At the destination, TCP **reassembles** the packets in the correct order.

Analogy: TCP is like a registered courier service. Every package is tracked and confirmed. If a package is lost, it is re-sent.

IP — Internet Protocol

IP handles the **addressing and routing** of packets.

- Each packet is labelled with the **source IP address** (where it came from) and the **destination IP address** (where it is going).
- Routers use these IP addresses to forward packets across the internet.

Analogy: IP is like writing an address on an envelope. Without an address, the postal system cannot route the envelope anywhere.

UDP — User Datagram Protocol

UDP is an alternative to TCP. It sends data **faster** but without guarantees.

- No confirmation that packets arrived.
- No re-sending lost packets.
- Much **faster** because there is no waiting for confirmations.

When is UDP used? When speed matters more than perfection — like video calls, live streaming, or online gaming. In a video call, it is better to skip one frame than to pause the entire call waiting for it.

TCP vs UDP

Feature	TCP	UDP
Reliability	High (confirms delivery)	Low (no confirmation)
Speed	Slower	Faster
Error checking	Yes	Minimal
Use case	Web browsing, email, file download	Video calls, gaming, live streaming

Grab a Partner — 3.8

Role Play — DNS in Action!

- **Person A** plays the role of a **web browser**.
- **Person B** plays the role of a **DNS server**.

Script:

- Person A (browser): "I need to find `www.pakschool.edu`. What is the IP address?"
- Person B (DNS server): "Let me check... The IP address is `103.45.22.11`."
- Person A (browser): "Thank you! I will now contact `103.45.22.11` to get the webpage."

Now switch roles. Then discuss: What would happen if the DNS server was unavailable? Could you still visit websites?

3.9 Network Security

The Hook: The Lock on the School Gate

Every school has a gate. At the gate, there is usually a guard. The guard checks who is entering — students and staff are allowed in, strangers without permission are stopped.

But what if the school had no gate and no guard? Anyone could walk in — to steal, to damage property, or to cause harm.

Computer networks are exactly like a school. They contain valuable information — personal data, financial records, private messages. Without security, anyone could enter and cause harm.

Network security is the gate and the guard for your computer network.

What is Network Security?

Network security refers to all the measures, tools, and practices used to **protect a computer network and its data** from unauthorised access, misuse, damage, or theft.

Why is Network Security Important?

There are four key reasons why we must protect networks:

1. Data Protection

Networks store enormous amounts of **sensitive information** — private messages, banking details, medical records, exam results, passwords.

If attackers access this data, they can steal identities, drain bank accounts, or blackmail people.

Example: If a hospital's network is breached, patient medical records (including diagnoses and treatments) can be stolen and misused.

2. Preventing Attacks

Networks face constant attacks from malicious (harmful) software and people who want to disrupt, damage, or take control of systems.

Common attacks include:

- **Malware:** Harmful software designed to damage or infiltrate a system.
- **Phishing:** Fake emails or websites that trick users into giving their passwords.
- **Ransomware:** Software that locks your files and demands money to unlock them.

Example: In 2017, a ransomware attack called "WannaCry" attacked the UK's National Health Service (NHS), locking doctors and nurses out of patient records for days.

3. Maintaining Privacy

Networks carry **private communications** — personal emails, private conversations, confidential documents. Network security ensures these stay private and are not read by unauthorised people.

4. Ensuring Availability

A network must be **available** (working and accessible) whenever authorised users need it. Attackers sometimes try to overwhelm networks with fake traffic, causing them to crash and become unavailable. This is called a **Denial of Service (DoS) attack**.

Example: A school's online exam portal must be available on exam day. A DoS attack could crash it just when 1,000 students are trying to log in.

Pause & Think — 3.9

Scenario: You receive an email that says: "Congratulations! You have won a prize. Click here and enter your school login and password to claim it."

Questions:

1. Is this email trustworthy? How do you know?
2. What type of attack is this?
3. What should you do?

Discuss with a partner before reading the next section.

3.10 Network Security Methods

The Hook: Three Layers of Security at a Bank

A well-designed bank uses multiple layers of security:

- **The front door** is locked and monitored (only authorised people enter).
- **The safe** uses a combination lock and scrambles everything inside (even if someone enters, the valuables are protected).
- **Security cameras** watch for suspicious behaviour (detecting threats quickly).

Three layers. Each one different. Together, they create strong security. Network security works the same way.

Security Method 1: Firewalls

What is a Firewall?

A **firewall** is a security system that **monitors and controls** all incoming and outgoing network traffic. It acts as a **filter** — allowing safe traffic to pass and blocking dangerous or unauthorised traffic.

Analogy: A firewall is like the security guard at the school gate. Every person (data packet) entering or leaving is checked. Those who belong are let through. Those who do not belong are blocked.

How Does a Firewall Work?

A firewall examines each data packet and checks it against a set of **rules**:

- **Is this packet coming from a trusted source?** → Allow.
- **Is this packet from an unknown or blocked source?** → Block.
- **Is this packet trying to access a restricted part of the network?** → Block.

Firewalls can be:

- **Hardware firewalls:** Physical devices installed between your network and the internet.
 - **Software firewalls:** Programs installed on your computer (Windows has a built-in software firewall).
-

Advantages of Firewalls

- ✗ • Blocks unauthorised access.
- Prevents many common attacks.
- Can be customised to allow or block specific types of traffic.

Disadvantages of Firewalls

- Cannot stop attacks that come from *inside* the network.
 - May block legitimate traffic if rules are set incorrectly.
 - Does not protect against viruses already inside the system.
-

Security Method 2: Encryption

What is Encryption?

Encryption is the process of **converting data into a secret code** so that only authorised people can read it.

Analogy: Imagine writing a letter in a secret language that only you and your friend know. Even if someone steals the letter, they cannot read it because they don't know the code. Encryption does this for digital data.

When data is encrypted:

- The original data is called **plaintext** (readable).
 - The encrypted data is called **ciphertext** (scrambled, unreadable).
 - A special **key** is needed to decrypt (unscramble) the ciphertext back into plaintext.
-

Where Do We See Encryption?

- **HTTPS websites:** When you visit a website with "https://" in the address, your connection is encrypted. Look for the padlock icon in the browser address bar.
 - **WhatsApp messages:** WhatsApp uses **end-to-end encryption**, meaning only you and the person you're messaging can read your messages — not even WhatsApp itself.
 - **Banking apps:** All transactions are encrypted to protect your financial data.
-

Advantages of Encryption

- Even if data is intercepted, it cannot be read without the key.

- Protects data both in transit (being sent) and at rest (stored on a device).
- Used on billions of devices worldwide, with well-tested methods.

Disadvantages of Encryption

- Requires processing power — can slow down devices slightly.
 - If the encryption key is lost, data may be unrecoverable.
 - Very sophisticated attackers may attempt to break encryption (though modern encryption is extremely strong).
-

Security Method 3: Antivirus Software

What is Antivirus Software?

Antivirus software is a program that **detects, prevents, and removes malicious software** (called malware) from your computer and network.

Types of malware antivirus software protects against:

- **Viruses:** Programs that attach to files and spread from device to device.
- **Worms:** Programs that spread over networks without needing to attach to files.
- **Trojans:** Programs that pretend to be useful software but secretly harm your system.
- **Ransomware:** Programs that lock your data and demand payment.
- **Spyware:** Programs that secretly record your activities (passwords, keystrokes, etc.).

Analogy: Antivirus software is like a **doctor** for your computer. It regularly examines your system, identifies infections (malware), and removes them before they can cause serious harm.

How Does Antivirus Software Work?

1. **Scanning:** It regularly scans all files on your device, looking for known patterns of malware (called **signatures**).
 2. **Real-time protection:** It monitors your device continuously, blocking suspicious programs before they can run.
 3. **Quarantine:** If it finds a dangerous file, it isolates it (quarantines it) so it cannot harm the rest of the system.
 4. **Updates:** Antivirus software regularly updates its database of known malware signatures to detect new threats.
-

Advantages of Antivirus Software

- ✗ • Protects against known malware efficiently.
- Runs automatically in the background.
- Regular updates help protect against new threats.

Disadvantages of Antivirus Software

- Cannot always detect brand-new malware (before the signature database is updated).
- Can slow down the computer slightly.
- Not 100% effective — must be combined with other security methods.

Security Methods Summary

Method	What it Does	Protects Against	Limitation
Firewall	Monitors and filters network traffic	Unauthorised access, external attacks	Does not stop internal threats or viruses
Encryption	Scrambles data so only authorised users can read it	Data theft during transmission	Key must be protected
Antivirus	Detects and removes malicious software	Viruses, worms, ransomware, spyware	May miss brand-new malware

Best Practice: Use all three methods together. A firewall alone, encryption alone, or antivirus alone is not enough. **Layered security** — multiple methods combined — is always stronger.

Pause & Think — 3.10

Scenario: Your school computer lab has 30 computers connected to the internet. One student accidentally downloads a file with a virus. The virus starts spreading through the network.

Challenge: Which security method would have:

1. **Prevented** the download from happening?
2. **Detected** the virus once it was on a computer?
3. **Protected** the data even if someone managed to steal it?

Write the name of the correct security method for each situation.

3.11 Types of Networks

The Hook: From Your Pocket to the Planet

Think about the range of your daily communications:

- **Your earbuds** connect to your phone via Bluetooth — a distance of about 1 metre.
- **Your phone** connects to your home Wi-Fi — a distance of about 20 metres.
- **Your school** connects computers across an entire building — a distance of about 100 metres.
- **Your city** has a municipal broadband network — a distance of perhaps 30 kilometres.
- **Your country** connects to the entire world — a distance of thousands of kilometres.

Every one of these connections is a *different type of network*. Networks are classified by their **size, range, and purpose**.

Network Type 1: Personal Area Network (PAN)

What is a PAN?

A **Personal Area Network (PAN)** is the **smallest type of network**. It connects personal devices over a very short distance — typically just a few metres — around a single person.

Think of it as: Your personal bubble of connectivity.

Common Technologies Used: Bluetooth, infrared (IR), USB.

Typical Range: Up to **10 metres**.

Examples of a PAN

- Your **phone** connected to **wireless earbuds** via Bluetooth.
 - Your **laptop** connected to your phone via Bluetooth to use the phone's internet connection (mobile hotspot).
 - Your **smartwatch** syncing data with your phone.
 - A **wireless keyboard and mouse** connected to your laptop.
-

Advantages of PAN

- × • Very simple and inexpensive to set up.
- No wires needed (for wireless PAN).
- Ideal for personal device synchronisation.

Disadvantages of PAN

- Very limited range (just a few metres).
- Can only connect a small number of devices.
- Not suitable for sharing resources across a large area.

Network Type 2: Local Area Network (LAN)

What is a LAN?

A **Local Area Network (LAN)** connects computers and devices within a **limited geographic area** — such as a single home, classroom, school building, or office.

Typical Range: Up to **1 kilometre** (within a building or campus).

Common Technologies Used: Ethernet cables, Wi-Fi.

Examples of a LAN

- All the computers in your school's computer lab connected to each other and to a shared printer.
- All devices in your home connected to your home Wi-Fi router.
- The computers in an office connected to a central server.

How to Set Up a Simple LAN (Step-by-Step)

Here is how a simple wired LAN is set up in a computer lab:

1. **Place a network switch** in a central location in the lab.
2. **Run an Ethernet cable** from the switch to each computer.
 - *What just happened?* You have physically connected all computers to the switch.
3. **Connect the switch to a router** using another Ethernet cable.
 - *What just happened?* The switch is now connected to the internet via the router.

4. **Turn on all devices.** They should automatically receive IP addresses from the DHCP server (usually in the router).
 5. **Test the connection** by trying to access the internet or ping another computer on the network.
-

Advantages of LAN

- Fast data transfer speeds (especially with Ethernet cables).
- × • Easy to share resources like printers and files.
- Relatively inexpensive to set up.
- Secure — usually contained within a building.

Disadvantages of LAN

- Limited to a small geographic area.
 - Requires physical cabling (for wired LANs).
 - If the central switch fails (in star topology), the whole network may go down.
-

Network Type 3: Metropolitan Area Network (MAN)

What is a MAN?

A **Metropolitan Area Network (MAN)** is a network that spans a **city or large campus**, connecting multiple LANs together.

Think of it as: A network big enough to connect different parts of a city — like different branches of a university, or different government offices across town.

Typical Range: Up to **50 kilometres**.

Examples of a MAN

- A university with multiple campuses across a city — all campuses connected into one network.
 - A city government connecting its offices, police stations, and hospitals across the city.
 - A cable TV provider connecting subscribers across a city.
-

Advantages of MAN

- ✗ • Covers a much larger area than a LAN.
- Allows organisations with multiple locations in a city to share resources.
- Faster than WAN (because it covers a smaller area).

Disadvantages of MAN

- More expensive to set up than a LAN.
- More complex to manage.
- Requires significant infrastructure (cables, equipment) across the city.

Network Type 4: Wide Area Network (WAN)

What is a WAN?

A **Wide Area Network (WAN)** covers a **large geographical area** — connecting multiple LANs and MANs together across cities, countries, or even continents.

The Internet is the largest WAN in the world — connecting billions of devices across every continent.

Typical Range: Thousands of kilometres — country-wide or global.

Examples of a WAN

- The **Internet** — the global network connecting billions of devices worldwide.
- A multinational company (like a bank or airline) connecting its offices in Karachi, Islamabad, London, and New York into one network.
- A government connecting all its provincial offices across a country.

Advantages of WAN

- Connects devices across enormous distances.
- Enables global communication and collaboration.
- Allows remote access to centralised resources.

Disadvantages of WAN

- Very expensive to set up and maintain.
- Slower than LAN or MAN (because data travels greater distances).
- More vulnerable to security threats (because the network is large and passes through many hands).
- Requires leasing lines from telecommunications companies, which is costly.

Security Tip: When connecting to a WAN (especially a public one), always use a **VPN (Virtual Private Network)** to encrypt your connection and protect your data.

Network Type 5: Campus Area Network (CAN)

What is a CAN?

A **Campus Area Network (CAN)** connects multiple LANs within a **limited geographical area** such as a university campus, a business park, a hospital complex, or a military base.

Think of it as: A MAN for a specific campus or grounds — larger than a single LAN (one building), but smaller than a full city MAN.

Typical Range: A few hundred metres to a few kilometres (within campus grounds).

Examples of a CAN

- A large university connecting its Library, Engineering Department, Business School, and Administration Block into one network — all on the same campus.
- A hospital complex connecting its Emergency Ward, Pharmacy, and Administration offices.
- A large technology company's headquarters connecting multiple buildings on the same grounds.

Advantages of CAN

- Connects multiple buildings on the same grounds efficiently.
- Faster than a MAN or WAN (shorter distances).
- More manageable than a MAN (limited to one campus).

Disadvantages of CAN

- Limited to a specific campus — does not extend to the wider city.
- Setting up cables between multiple buildings can be expensive.

Network Types Summary Table

Network Type	Full Name	Coverage Area	Example
PAN	Personal Area Network	A few metres	Bluetooth earbuds + phone
LAN	Local Area Network	A building or floor	School computer lab
CAN	Campus Area Network	A campus or complex	University campus
MAN	Metropolitan Area Network	A city	City government offices
WAN	Wide Area Network	Countries / global	The Internet

Pause & Think — 3.11

Scenario: Here are four situations. Identify which type of network is being described (PAN, LAN, CAN, MAN, or WAN):

1. A student's wireless earphones are connected to their phone while walking in the school corridor.
2. All computers in a school lab can print to a shared printer.
3. A university with five buildings on one campus shares a single internet connection across all buildings.
4. An international airline company connects its ticket offices in Lahore, London, Dubai, and New York.
5. A city's metro train system uses one connected network for all stations across the city.

3.12 Real-World Applications of Computer Networks

The Hook: A World Without Networks

Imagine waking up tomorrow and every computer network in the world has stopped working. No internet. No Wi-Fi. No mobile data.

- Hospitals cannot access patient records. Surgeries are cancelled.
- Banks cannot process transactions. ATMs don't work.
- Schools cannot deliver online lessons. Libraries cannot be searched.
- Businesses cannot communicate with each other. Deliveries stop.

It would be a catastrophe. This thought experiment reveals something important: computer networks are not just *useful* — they are the **invisible infrastructure** on which modern life depends. Let's explore where we see them most.

Application 1: Business

How Networks Transform Business

In the modern business world, networks are as essential as electricity. They enable:

Resource Sharing Across Branches

A company with offices in multiple cities can connect all its computers into one network. Employees in different cities can access the same files, databases, and software — as if they were in the same room.

Example: A clothing company in Lahore has branches in Karachi and Islamabad. Through their WAN, managers in all three cities can access the same inventory database in real time.

The Intranet: A Private Internet

Many companies build an **Intranet** — a private network that works like the internet but is only accessible to employees within the organisation.

An intranet may contain:

- Company policies and manuals
- Internal job postings
- Employee directories
- Internal news and announcements
- Shared project files and tools

Think of it as: The internet, but with a locked door that only employees can open.

Key difference from the internet:

- The **Internet** is public — anyone can access it.
 - An **Intranet** is private — only people inside the organisation can access it.
-

Video Conferencing and Remote Work

Networks allow employees to work from home and still collaborate with colleagues thousands of kilometres away using video conferencing tools like:

- **Zoom**
- **Microsoft Teams**
- **Google Meet**

Example: During the COVID-19 pandemic, millions of employees worldwide used network-based video conferencing to continue working from home. Without computer networks, this would have been impossible.

Network Benefits for Business

- Reduces costs (shared resources, less travel)
 - Increases productivity (real-time collaboration)
 - Enables remote work and flexible working hours
 - Allows companies to operate globally
-

Application 2: Education

How Networks Transform Education

Computer networks have fundamentally changed how teaching and learning happen — especially in the last decade.

Learning Management Systems (LMS)

Universities and schools now use **Learning Management Systems (LMS)** — software platforms that allow teachers to:

- Upload course materials (notes, videos, assignments)
- Give and receive assignments online
- Grade students and provide feedback
- Communicate with students through discussion forums

Popular LMS platforms include:

- **Blackboard**
- **Moodle** (often used in Pakistani universities)
- **Google Classroom**
- **Canvas**

Example: A teacher uploads a video lecture on Tuesday night. Students access it from home before the Wednesday class. In class, they discuss the video instead of listening to a lecture. This is called **flipped learning** — only possible because of networks.

Online Libraries and Research Databases

Networks give students access to **millions of books, research papers, and journals** that would be impossible to store in a physical library.

Example: A student in a small town in rural Pakistan can access research papers from Harvard University's library through the internet — for free, if the university subscribes to the service.

Virtual Classrooms

Networks enable **virtual classrooms** — fully online classes where students and teachers meet in real time through video, audio, and chat.

Example: During examinations or emergencies, Pakistani universities shifted entirely to online learning using platforms like Zoom. Students attended live lectures, submitted assignments, and even gave presentations — all through the network.

Network Benefits for Education

- Provides access to knowledge regardless of geographic location
 - Enables personalised learning at each student's own pace
 - Connects students with experts and resources worldwide
 - Reduces the cost of education (no need to physically travel)
-

Application 3: Healthcare

How Networks Transform Healthcare

Healthcare is perhaps the most critical application of computer networks. Mistakes in healthcare can cost lives — and networks help prevent those mistakes.

Electronic Health Records (EHR)

In traditional hospitals, patient records were written on paper and stored in filing cabinets. Finding a patient's history could take hours. Sharing it with another doctor in a different hospital was almost impossible.

Today, hospitals use **Electronic Health Records (EHR)** — digital versions of patient records stored in networked databases.

With EHR:

- A doctor can access a patient's complete medical history in seconds.
- If a patient visits a different hospital, their records can be shared immediately.
- Prescription errors are reduced (the system can warn if two medications are dangerous together).
- Patient data is backed up automatically — it cannot be lost in a fire or flood.

Example: A patient arrives unconscious at an emergency room. The doctor cannot speak to them. Using the EHR network, the doctor accesses the patient's history in seconds — discovering they are allergic to a common painkiller. This information could save their life.

Telemedicine

Telemedicine allows doctors to consult with patients **remotely** — through video calls, messages, or even by analysing medical images sent over the network.

Example: A patient in a remote village in northern Pakistan cannot easily travel to a specialist hospital in Lahore. Using telemedicine, the specialist examines the patient via a video call and reviews their medical images sent through the network — providing expert care without the patient leaving their village.

Medical Research Networks

Hospitals and universities worldwide connect their research data through networks, allowing scientists to:

- Share medical data across countries
- Collaborate on finding cures for diseases
- Run large-scale clinical trials with patients in multiple countries

Example: During the COVID-19 pandemic, researchers worldwide shared genomic data of the virus through international networks in real time. This allowed scientists in different countries to develop vaccines simultaneously — dramatically speeding up the response.

Network Benefits for Healthcare

- Faster access to patient information saves lives
 - Reduces medical errors through shared, complete records
 - Extends expert healthcare to remote areas via telemedicine
 - Accelerates medical research through global collaboration
-

Final Pause & Think — 3.12

Big Picture Challenge: Think about your own daily life for 24 hours — from when you wake up to when you go to sleep.

Challenge: Write down **5 moments** in your day when you use (or benefit from) a computer network — even indirectly. Try to identify which *type* of network is involved (LAN, WAN, etc.) and which *protocol* or *security method* might be working behind the scenes.

Example start: "7:00 AM — I check WhatsApp messages. This uses a WAN (the internet), the TCP/IP protocol, and HTTPS encryption."

Share your list with a partner and compare. Did you think of moments the other person missed?

Chapter Summary

Congratulations! You have completed the entire Computer Networks chapter. Here is a quick review of everything you have learned:

Topic	Key Takeaway
Network as a System	A network connects devices to share resources, communicate, and collaborate.
Data Communication	Five components: Sender, Receiver, Message, Protocol, Medium.
Switch	Connects devices within one network using MAC addresses.
Router	Connects different networks using IP addresses.
Access Point	Connects wireless devices to a wired network.
Bus Topology	All devices on one cable. Simple, but one failure = whole network down.
Star Topology	All devices connected to a central switch. Best for most environments.
Ring Topology	Devices in a circle. One failure affects the whole ring.
Mesh Topology	Every device connected to every other. Most reliable, most expensive.
Simplex	One-way data flow only (e.g., keyboard to computer).
Half-Duplex	Two-way, but not simultaneously (e.g., walkie-talkie).
Full-Duplex	Two-way simultaneously (e.g., phone call).
OSI Model	7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
IPv4	32-bit addresses (~4.3 billion). Format: 192.168.1.1
IPv6	128-bit addresses (~340 undecillion). Created to replace IPv4.
DNS	Translates website names to IP addresses.
DHCP	Automatically assigns IP addresses to devices.
TCP/IP	The core protocol suite of the internet. TCP = reliable. IP = addressing.
Network Security	Protects data, prevents attacks, maintains privacy and availability.
Firewall	Filters incoming/outgoing traffic based on rules.
Encryption	Scrambles data so only authorised users can read it.
Antivirus	Detects and removes malware.
PAN	Personal devices, a few metres (e.g., Bluetooth).
LAN	Within a building (e.g., school lab).

Topic	Key Takeaway
CAN	Within a campus (e.g., university grounds).
MAN	Within a city (e.g., city government offices).
WAN	Across countries/globe (e.g., the Internet).
Business	Networks enable resource sharing, intranets, and remote collaboration.
Education	Networks power LMS platforms, virtual classrooms, and online libraries.
Healthcare	Networks enable EHR systems, telemedicine, and medical research.

Final Encouragement

You have just explored one of the most important technologies ever created by humans. Every concept in this chapter — from switching to encryption to IPv6 — was invented by real people who also started out not knowing these things. Some of them were students just like you.

The internet, which connects billions of people today, began as a small network that crashed after sending just two letters.

Everything great begins small. You have taken a big step today.

End of Chapter 3: Computer Networks